

2026-10-06	1.0	
2026-10-06	1.1	

data/operation/ 13 1,572,894 2025-01-19 19:10:56 2025-09-11
18:07:11 CK7

□ □ □ □

- 25°C **25.63°C** 45°C **45.94°C** **0.63°C** **0.94°C**
- **1.14°C** **0.66°C** **0.48°C**
- **0**
- **67** **0.00426%** 67

□ □ □ □ □ □ □ □

chamber_temperature_C	25°C 45°C
temp_mean_C	
temp_min_C	
temp_max_C	

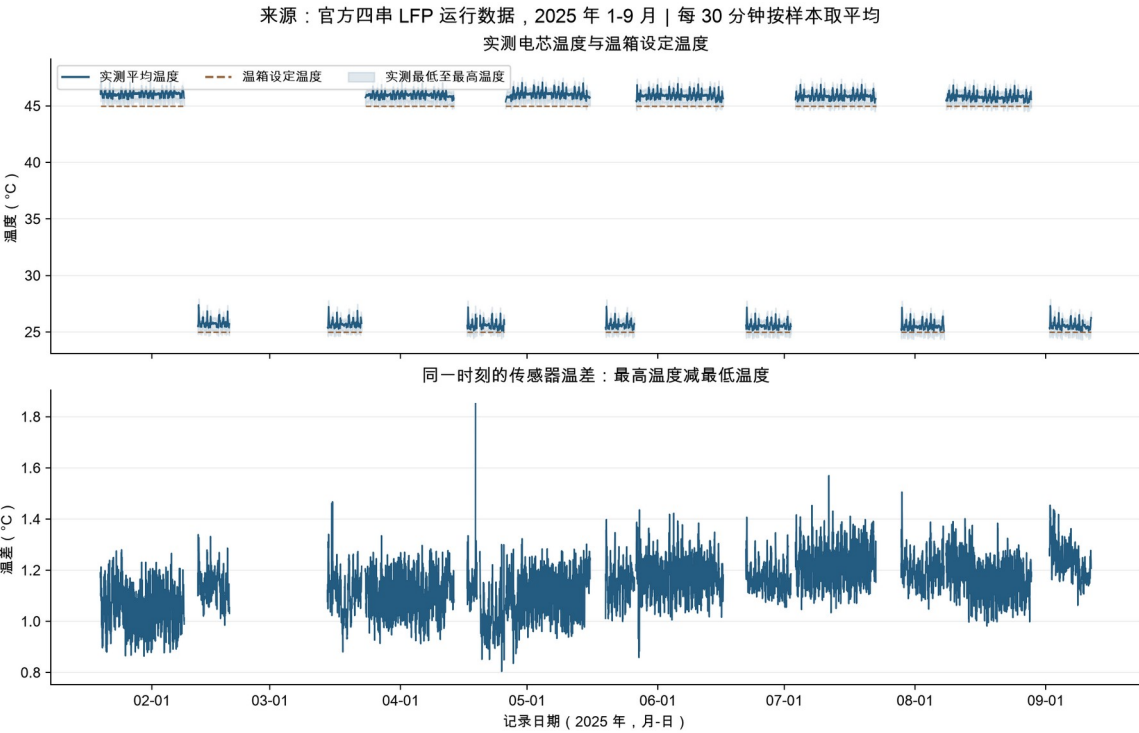
“ ”

□□□□□□□□□□

25℃ **529,012** 45℃ **1,043,882** “90% ” 5 95

	25℃	45℃
	25.627℃	45.939℃
	+0.627℃	+0.939℃
	+0.56℃	+0.91℃
90%	+0.19 +1.31℃	+0.50 +1.51℃
	24.89 27.49℃	44.10 47.61℃
	99.86%	99.995%

SOC 25℃ +0.729 +0.500℃ 45℃ +1.032 +0.827℃ >0.5 A <-0.5 A



1 2025 1 9 30

45℃ 1 +1.071℃ 13 +0.789℃

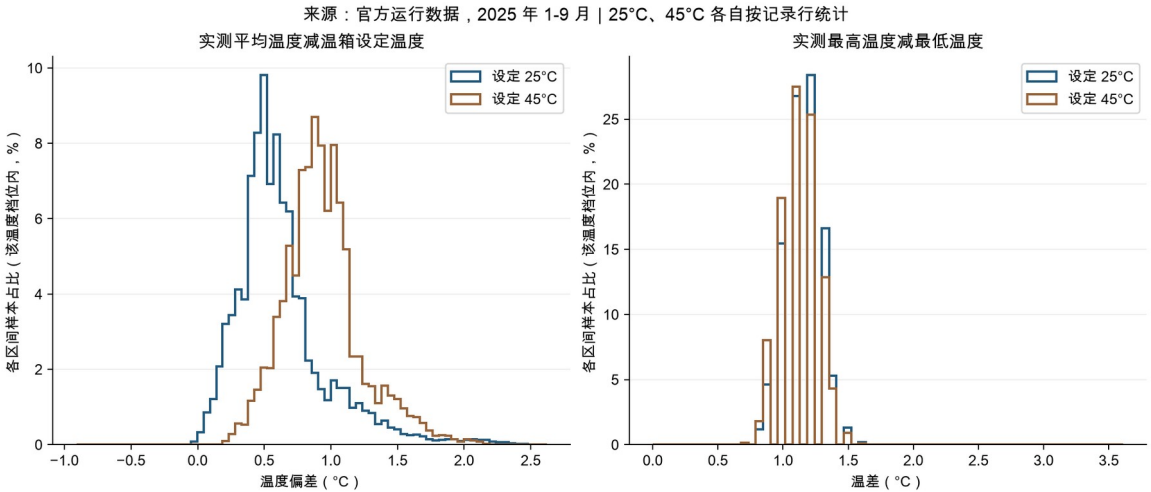
□ □ □ □ □ □ □ □ □ □

	25°C	45°C
	24.949°C	45.287°C
	25.627°C	45.939°C
	26.104°C	46.415°C
	0.678°C	0.652°C
	0.477°C	0.476°C
	1.155°C	1.127°C
90%	0.90 1.40°C	0.90 1.40°C
	2.20°C	3.60°C

45°C1.137°C1990.81.5°C

98%

0.092°C



225°C 45°C

□ □ □ □ □ □ □ □ □ □

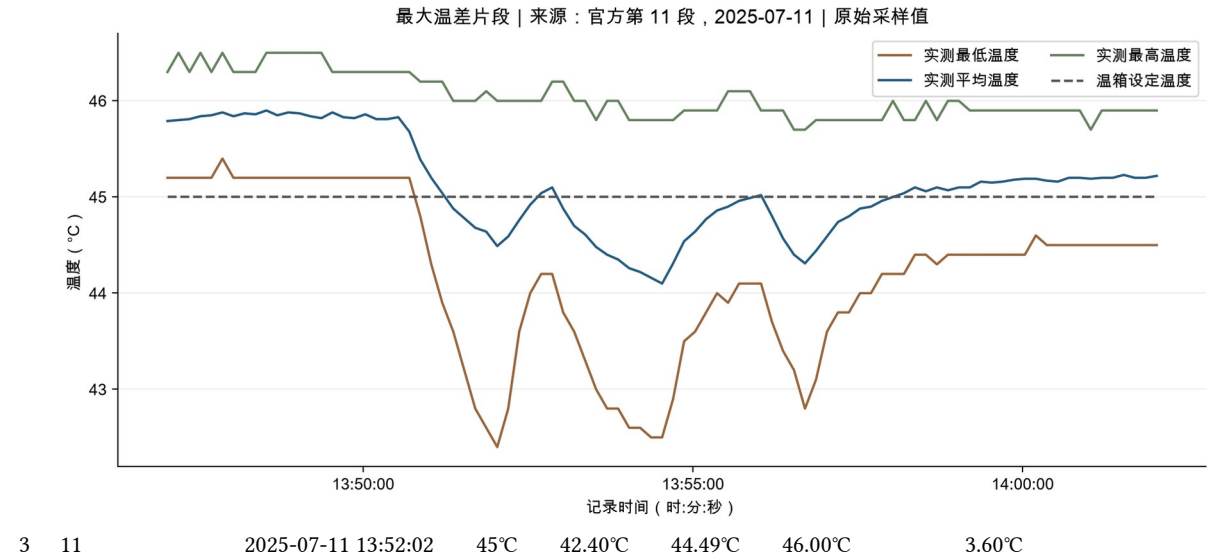
24.89 47.61°C 23.9 47.0°C 25.2 48.3°C

□□□□□□□□

6,291,576

NaN	0
	0
	0
	0
25℃ 45℃	0
≤ ≤	0

- **Δ2℃ 63** 0.00401% >3℃ 18
 - **Δ0.5℃ 9** >0 ≤30 10 0.8℃
- 5 **67** **0.00426%** >1℃ >2℃ 0 67 1 22 10 4 18 8
- 7 11 31 7 14 18



□□□□□□□□□□

3 2025 2 19 3 11 **checkup**
 >30 6 20.81 10 **20.79** 59,909 16
38 3

1	01-24 02:56:09	01-24 02:56:41	32
6	04-18 18:27:58	04-18 22:06:45	13,127
6	04-19 02:12:15	04-19 18:50:44	59,909
6	04-25 06:57:19	04-25 06:57:55	36
13	08-28 04:44:00	08-28 04:44:50	50
14	09-04 22:21:44	09-04 22:50:57	1,753

181 181 119 0.20°C

CK0

□□□□□□□□□□

0< ≤30
<0.00013°C
CK7 CK

□□□□□□□□□□

-
- 67
- CK1 CK7

□ □ □ □ □ □ □

“Arena 2 ”

Word

67

13 SHA256

2026 10 6

1,572,894

63+9-5=67

19911CK0

data/operation/131,572,89420251

□□□□

725°C12.2→13.9V

94.99Ah89.95Ah5.31%5.27%5.39%

SOH

□□□□□□□□

dataset original/

charge_Ah_cum / discharge_Ah_cum	Ah			
CK0 discharged_Ah	Ah	100.412 Ah	100.41 Ah	
SOH	5.1 A	11.2 V	SOH =	÷ 102 Ah × 100% CK0 98.44%

102 Ah102 Ah11.473 V11.2 V0

CK1–CK7

□□□□□□□□□□

16,585.075 Ah15,214.406 Ah3

checkup

2025-01-20 01:06:0901:36:08181–20.4 A10.19398 Ahcharge_Ah_cum10.200

Ah discharge_Ah_cum0 Ah

□□□□□□□□

|I|>0.5 A 15 30

≥13.95 V >15 A 41 ≥13.3 V 33 13.3 V 8
41 10

		V	V	h
	33	13.407–13.439	14.008–14.010	20.637–21.758
	8	11.617–12.037	14.009–14.010	91.975–100.143
20.4 A	41	13.365–13.602	13.160–13.226	10.194–10.200
≥15 Ah	444	13.065–13.400	12.846–13.305	15.618–50.232

33 13.347–13.363 V 16 mV 13.407–13.439 V 8
11.570–11.953 V 383 mV SOC
41 14.008–14.010 V 2 mV 13.979–13.987 V

□□□□□□□□□□

33 28 3 3

		3 h	3 h	
25°C	9	21.445	20.945	–2.33%
45°C	19	21.407	20.703	–3.29%

25°C 2 16 3 19 4 22 8 2 8 7 9 6 26.333°C 26.171°C 45°C
1 24 1 29 2 3 8 17 8 22 8 27 46.458°C 46.203°C
–2.33% –3.29%

□ □ □ □ □ □ □ □ □ □ □ □

8 7 25℃ 45℃ 12.2→13.9 V

26.33-26.59℃ 25℃ 20.2-20.6 A 20.4 A

		h	℃
2025-02-11	2	94.99	26.59
2025-03-14	4	93.61	26.47
2025-04-16	6	92.23	26.35
2025-05-19	8	91.73	26.40
2025-06-21	10	91.17	26.37
2025-07-28	12	90.61	26.33
2025-09-01	14	89.95	26.45

7 94.99494→89.95058 Ah 5.04437 Ah 5.31% 2025-02-11 2025-09-01 3 3 93.61225→90.61169 Ah 3.21%

□ □ □ □ □ □ □ □ □ □

V	h	h	
12.2→13.9	94.99	89.95	-5.31%
12.4→13.9	94.33	89.37	-5.27%
12.6→13.9	93.42	88.50	-5.27%
13.0→13.9	90.60	85.71	-5.39%

7 -5.27% -5.39%

SOC SOH

41	-20.4 A	29.983-30.000	10.194-10.200 Ah
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		V	V
25°C	16	13.365–13.411	13.160–13.171
45°C	25	13.462–13.602	13.212–13.226

444

19	45°C	3	3	13.5→13.9 V	-4.51%	13.6→13.9 V	-12.44%
13.7→13.9 V	-4.67%	13.8→13.9 V	-0.02%				

SOC

$$\Sigma[(I_{+} + I_{-}) \times \Delta t] / 7200 \quad A$$

“20.4 A×”	0.00136%	CK7
9	8	

SOH

□ □ □ □ □ □ □ □

“Arena 2 ”

CK0

41

LFP

		step_capacity_Ah	2	100 A	2.5 V	105.1542 Ah
105.1702 Ah				5.1 A	11.2 V	
2026	10	6	41			13.4 V

Word “ 1” “ 2”

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